ONE-PAGE BRIEF

Post-quantum cryptography review: trustwallet-wallet-core-en @ adf80f8bac98

Client: Trust Wallet — public repository (passive review) · Source: github.com/trustwallet/wallet-core, commit `adf80f8bac98` · Reviewed: 2026-11-04 19:39 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY
Level 0 of 4
no post-quantum cryptography

FINDINGS
0 Critical **0** High **10** Medium
1 Low **13** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)
Key establishment: 10 + 5 − 8 = 7 → **too late**
Signatures: 10 + 5 − 8 = 7 → **too late**

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

WHAT MATTERS MOST

Medium	QSI-2026-12926	**Transaction signing for 43 chain(s) relies on elliptic-curve signatures** confirmed by the analyst What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains...
Medium	QSI-2026-12916	**Quantum-vulnerable ECDSA: transaction and message signing (84 file(s))** confirmed by the analyst What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains...
Medium	QSI-2026-12917	**Quantum-vulnerable Ed25519: transaction and message signing (38 file(s))** confirmed by the analyst What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains...
Medium	QSI-2026-12919	**Quantum-vulnerable Schnorr (BIP-340): transaction and message signing (16 file(s))** confirmed by the analyst What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains...

7 more in the full report.

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium transaction and message signing	ECDSA, Ed25519, Schnorr (BIP-340), BLS12-381 ...	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum...
Medium authentication and identity (certificates, tokens, device attestation)	ECDSA	Keep credentials short-lived and plan ML-DSA certificates, tokens and keys as the ecosystem supports them
Medium role not determined	Ed25519, ECDSA, Schnorr (BIP-340), X25519 ...	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)...

KNOWN VULNERABLE COMPONENTS

Advisories for the versions in use: 5; unlikely to apply to this build: 5 (a Cargo feature that is off, affected functions not named in the code, or a platform that is not built); to review: 0.

Analyst review: 13 confirmed, 11 not applicable, 0 dismissed as false positives. (QSI review, 2026-10-28)

Static review of source code and manifests; files analysed: 1695. No requests were sent to the owner's systems.

Full report, content SHA-256: `f42e37d97d9fb8678004e87e0803924e2fc2f3508ba33d6fd7a8da749ec3f97b`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.